Comparative Study of PID and State-Space Pole Placement Control for a Differential Drive Mobile Robot

UNIVERSITY OF DEBRECEN • FACULTY OF ENGINEERING • DEPARTMENT OF MECHATRONICS

Project of Mechatronics

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SECTION 1 • PRESENTATION OUTLINE

Content of the Presentation

01

Problem & Task

Simulation-to-reality gap in mobile robotics.

02

Applied Principles

3-DoF kinematics, PID, State-Space observer.

03

Physical Solution

Simscape Multibody, DC motors, H-bridge.

04

Results & Future

Figure-8 metrics, conclusions, hardware plan.

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Ideal Theory vs. Physical Reality

Ideal Kinematic Simulation

- **Pure Rolling Contact:** Assumes zero longitudinal or lateral wheel slip.
- **Zero Inertial Delay:** Neglects chassis mass and wheel rotational inertia.
- **Instant Actuation:** Commands wheel velocities without electrical lag.
- **PID Bias:** Classical PID appears stable under idealized conditions.

Real Mechatronic Hardware

- **Multibody Contact Forces:** Tire stiffness, damping, ground friction slip.
- **Electromechanical Lag:** Cascaded DC motor torque and rotor inertia.
- **Power Electronics Limits:** H-bridge PWM switching and battery voltage drop.
- **Coupled Instability:** Unmodeled delays cause severe trajectory chatter.

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Chronological Engineering Methodology

- **Stage 1 — Mathematical Modeling:** Formulate 3-DoF differential drive kinematics and validate in Simulink .
- **Stage 2 — Ideal Control Design:** Test decoupled parallel PID versus State-Space pole placement with Luenberger observer .
- **Stage 3 — 3D Simscape Multibody:** Add rigid-body inertia, passive caster stability, revolute joints, and wheel-ground friction .
- **Stage 4 — Cascaded Motor Loop:** Couple outer path controllers with tuned inner-loop DC motor velocity PID controllers.
- **Stage 5 — Hardware Limitations:** Integrate non-ideal battery and H-bridge PWM drivers over a 30 s figure-8 track.

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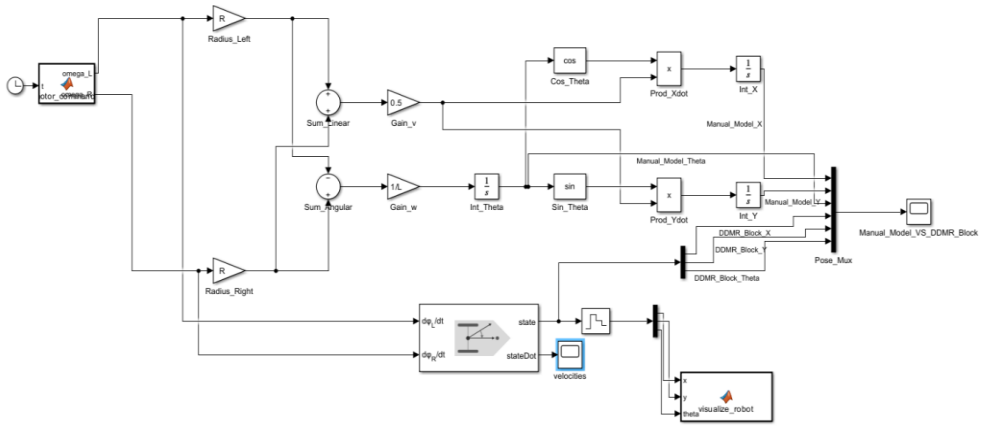
Step 1: 3-DoF Kinematics & Validation

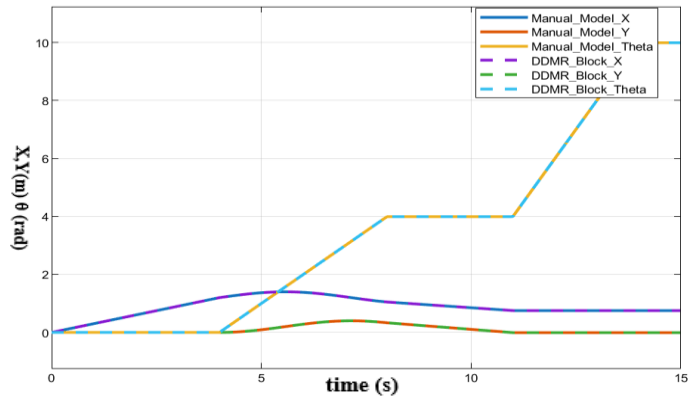
Kinematic Equations

$$v = \frac{R(\omega_R + \omega_L)}{2} \qquad \omega = \frac{R(\omega_R - \omega_L)}{L}$$

$$\dot{x} = v\cos(\theta) \qquad \dot{y} = v\sin(\theta) \qquad \dot{\theta} = \omega$$

$$\omega_{R,L}(t) = \frac{v(t) \pm \frac{L}{2}\omega(t)}{R}$$



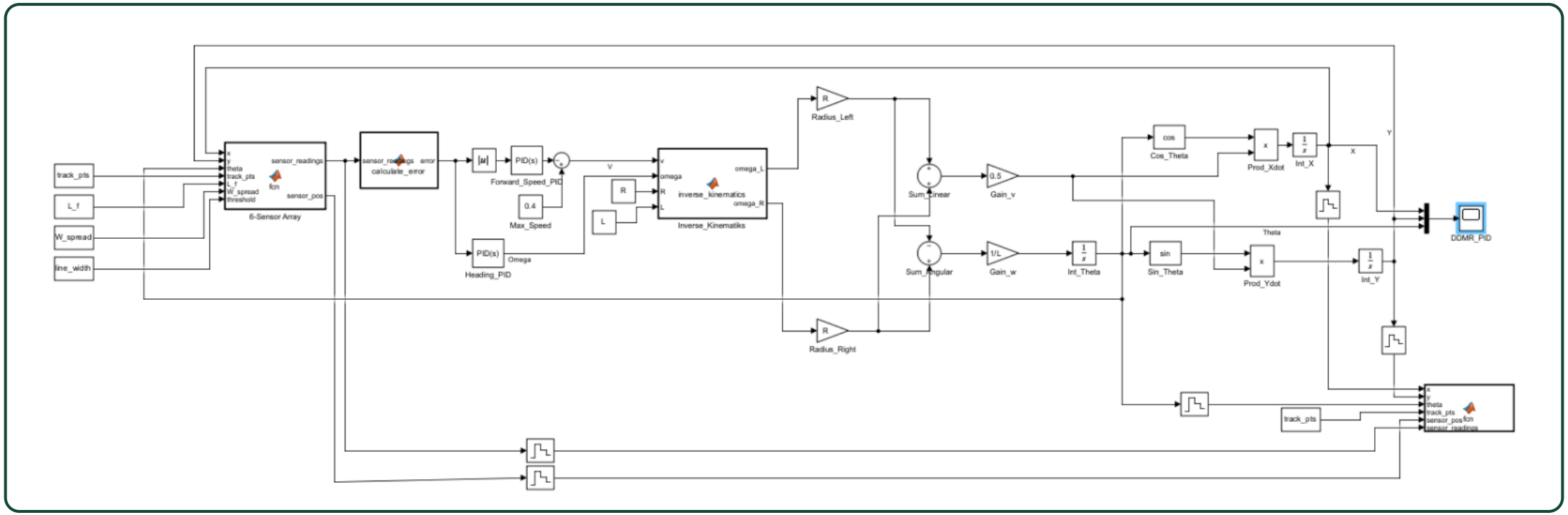


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Step 2A: Parallel PID Controller Design

$$\omega(t) = K_{p,\omega}\, e(t) + K_{i,\omega} \int e(\tau)\, d\tau + K_{d,\omega} \frac{de(t)}{dt} \qquad v(t) = 0.4 - \mathrm{PID}_v(|e(t)|)$$

Braking PID: (5, 0, 0) • Heading PID: (50, 0.001, 0)



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Step 2B: State-Space & Luenberger Observer

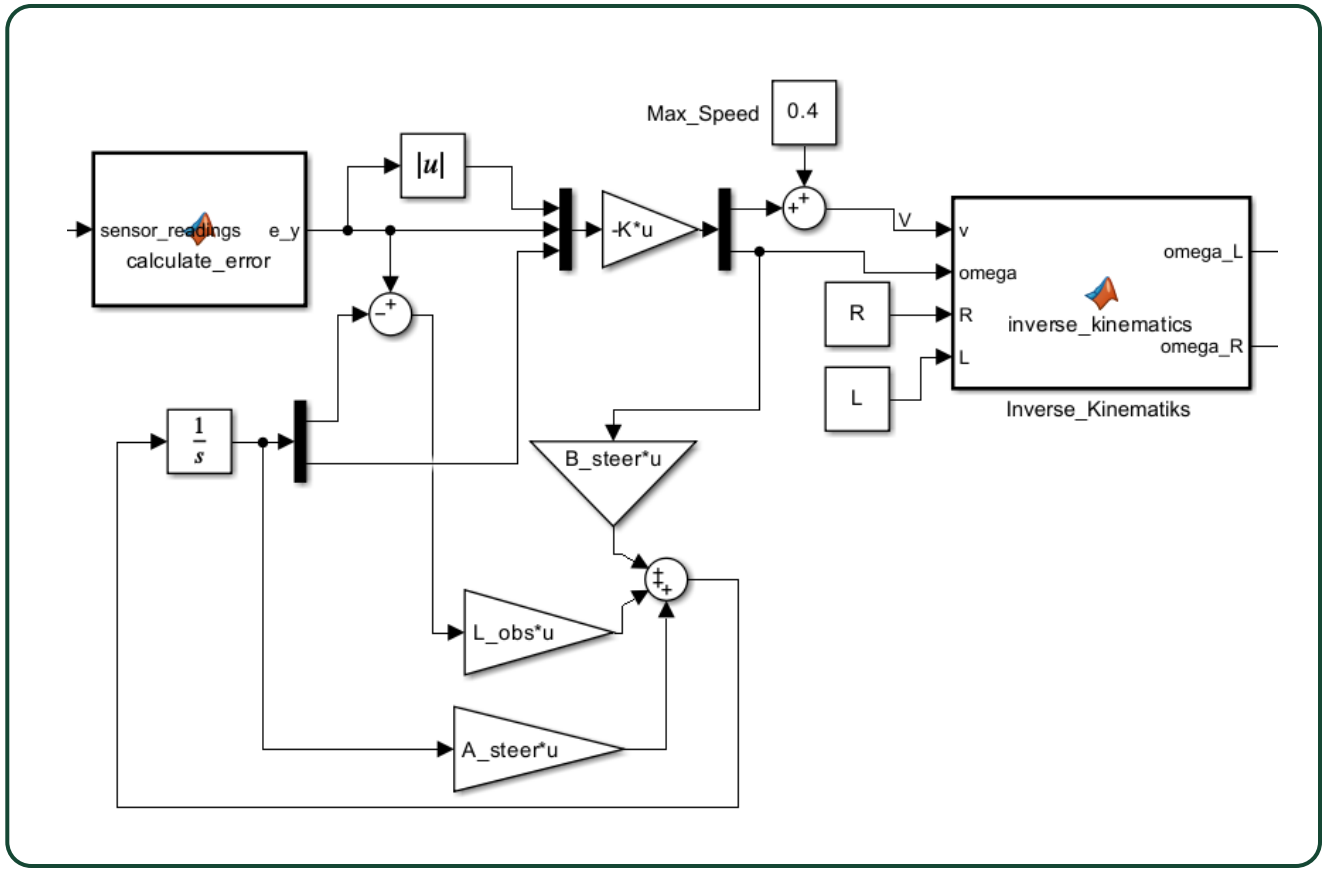
Observer & Pole Placement

$$\dot{\hat{x}}_{steer} = A_{steer}\,\hat{x}_{steer} + B_{steer}\,\omega + L_{obs}\,(y - \hat{y})$$

$$v(t) = V_{ref} - K_{brake}\,|\,e_y(t)\,|$$

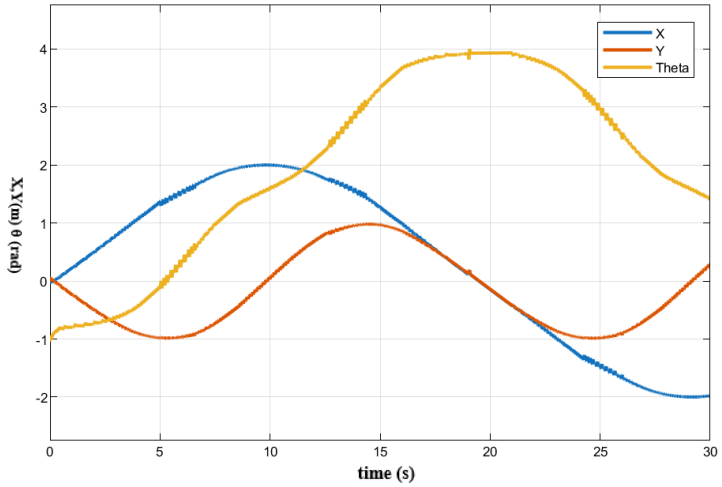
$$\omega(t) = -K_{steer,1}\,\hat{e}_y(t) - K_{steer,2}\,\hat{e}_\theta(t)$$

Initial Chosen Poles: [-0.9, -1.89]



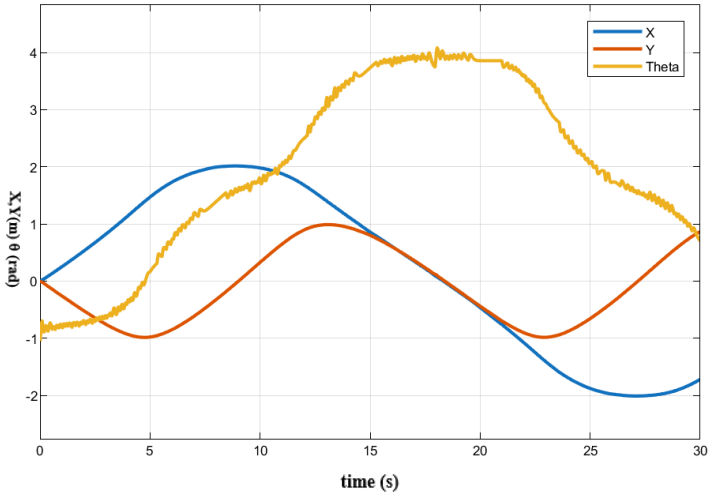
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Basic Math Model Graphs: PID vs. State-Space



Ideal PID Response

Follows crossover geometry with high-frequency micro-chatter.

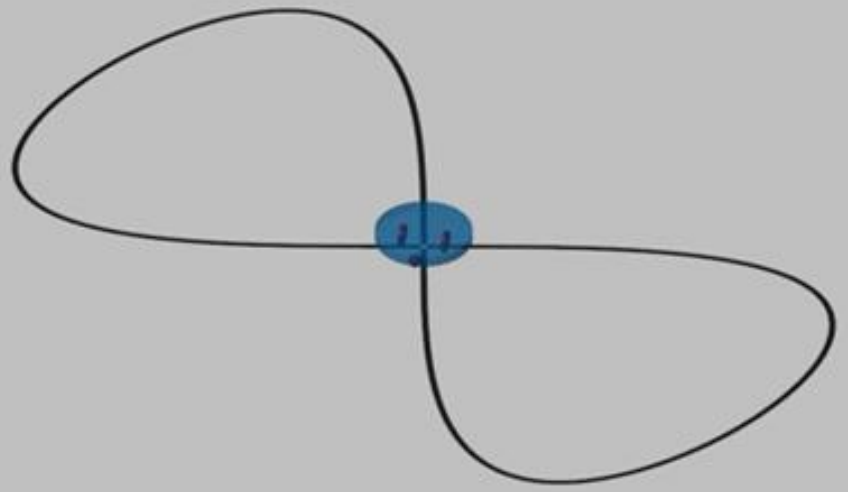


Ideal State-Space Response

Completes figure-8 path with slight heading wobble on turns.

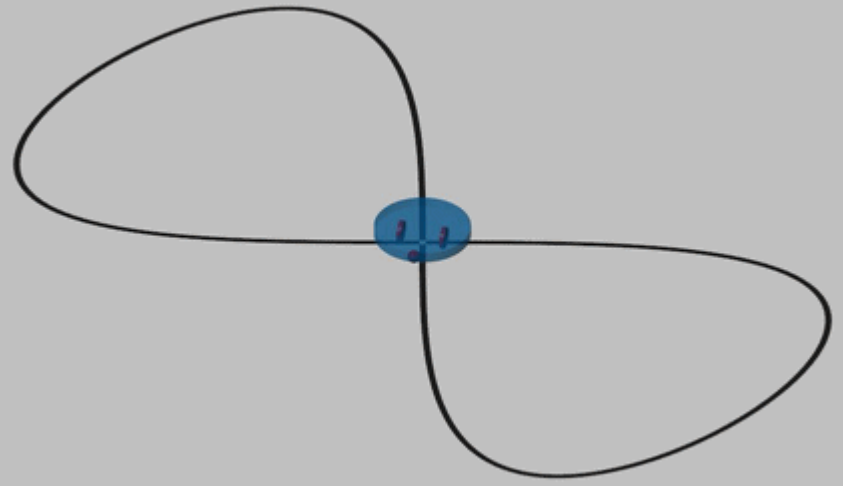
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Video Demonstration: Basic Math Models



PID Mathematical Model

Pure kinematic tracking at 0.40 m/s nominal velocity.

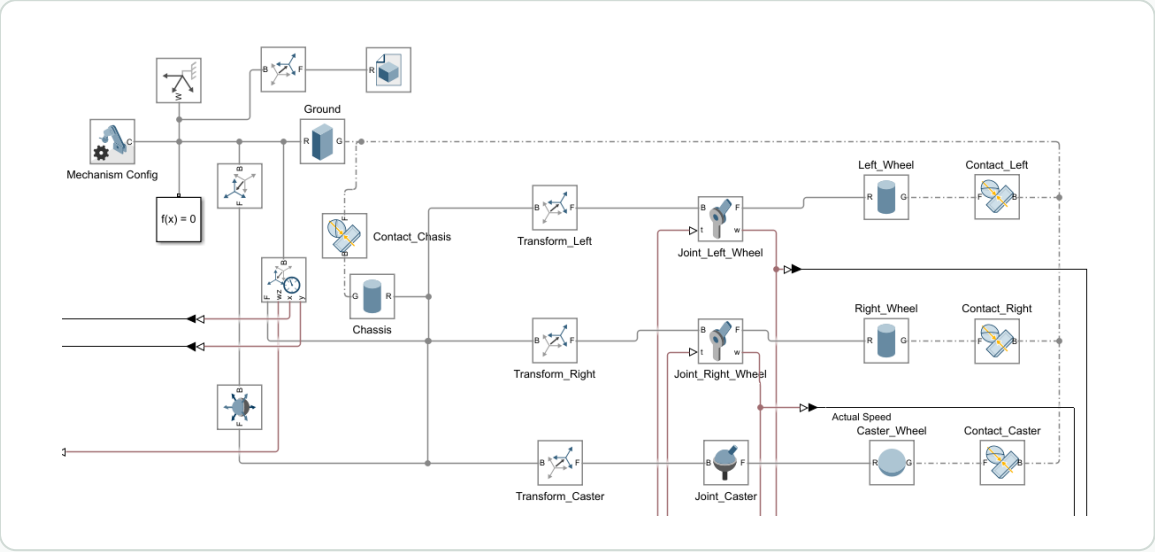


State-Space Mathematical Model

Pole placement with Luenberger observer state feedback.

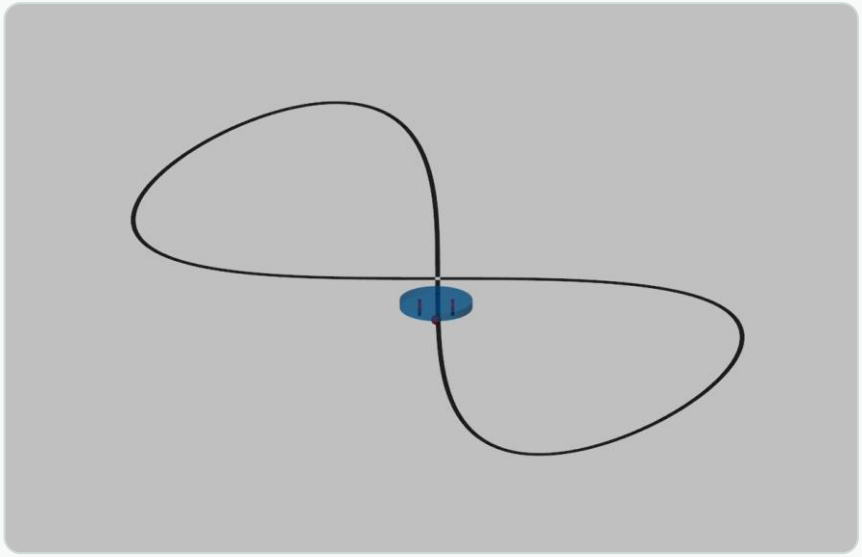
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Step 3: 3D Simscape Multibody Model



Multibody Plant Topology

Revolute drive joints, spherical caster, spatial contact forces.

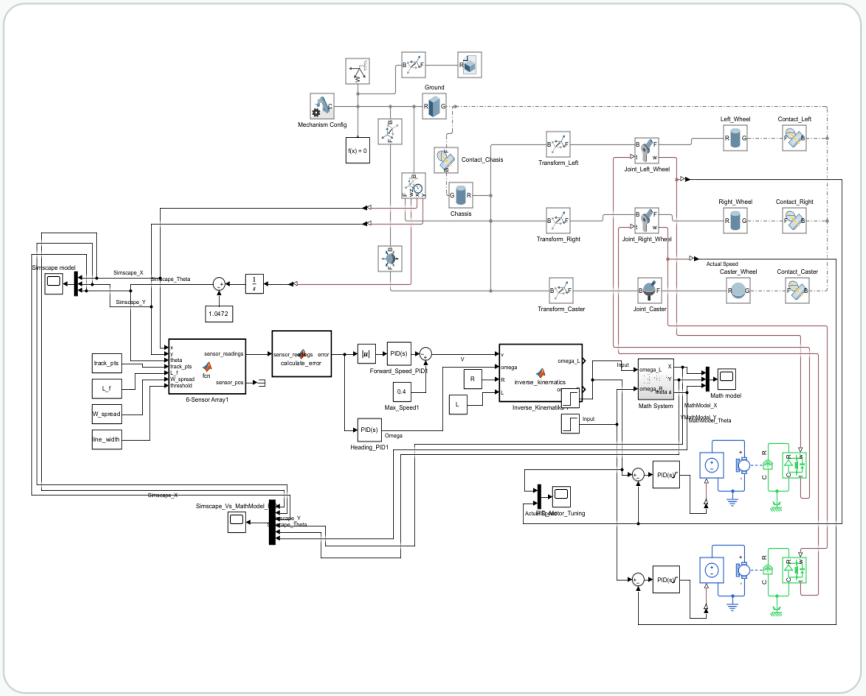


3D Physical Visualization

Cylindrical chassis with two drive wheels and passive caster.

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Step 4: Cascaded Motor Models & Tuning



Cascaded Models

Outer path loop drives inner wheel PIDs.

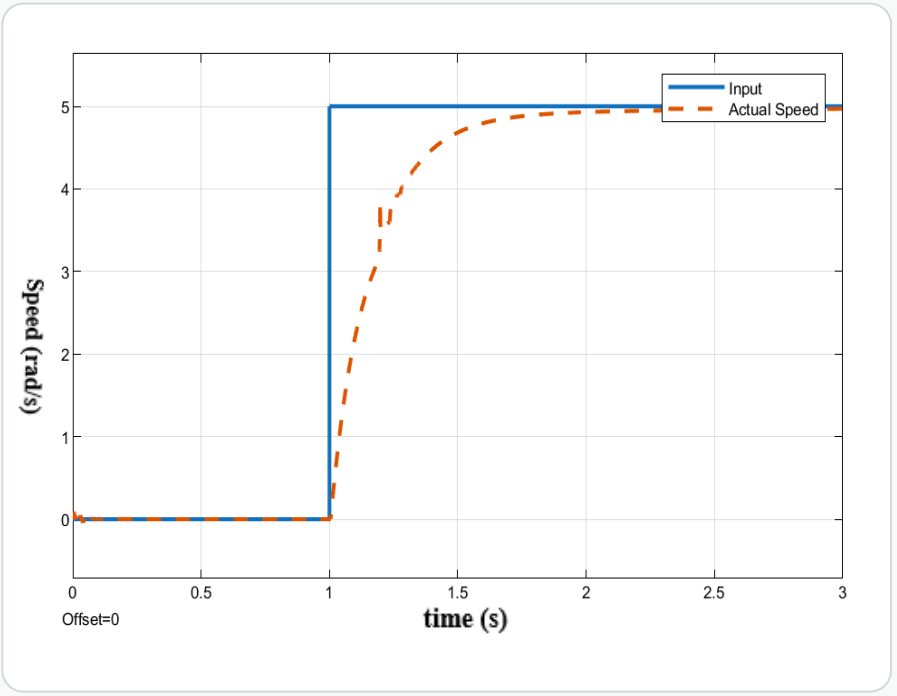
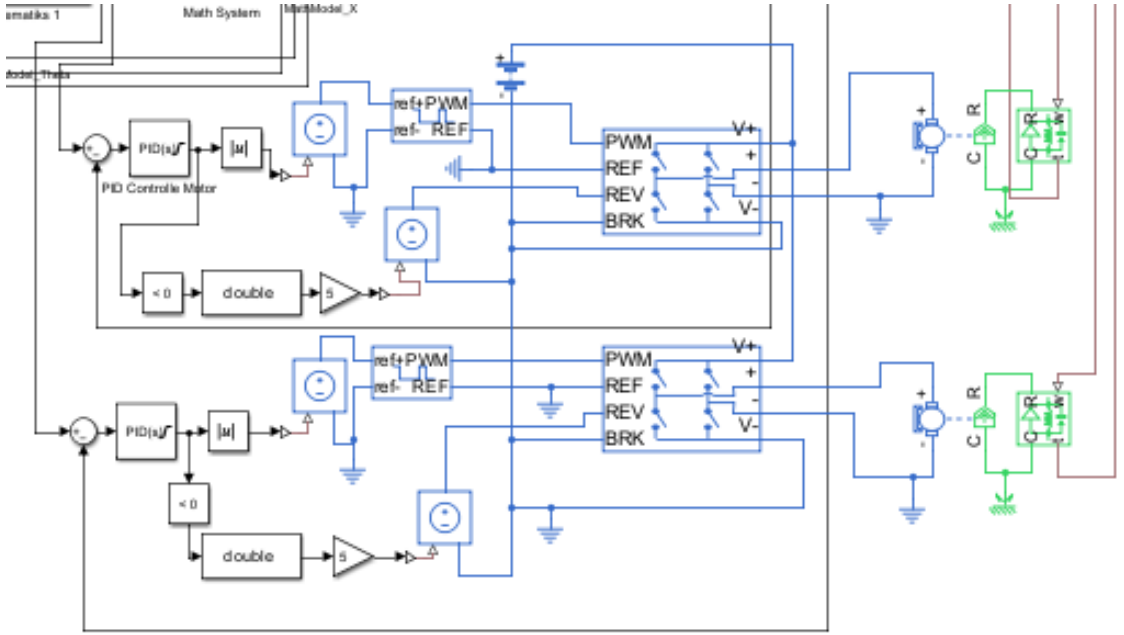


Figure 10: Motor Step

Fast 5 rad/s settling without overshoot.

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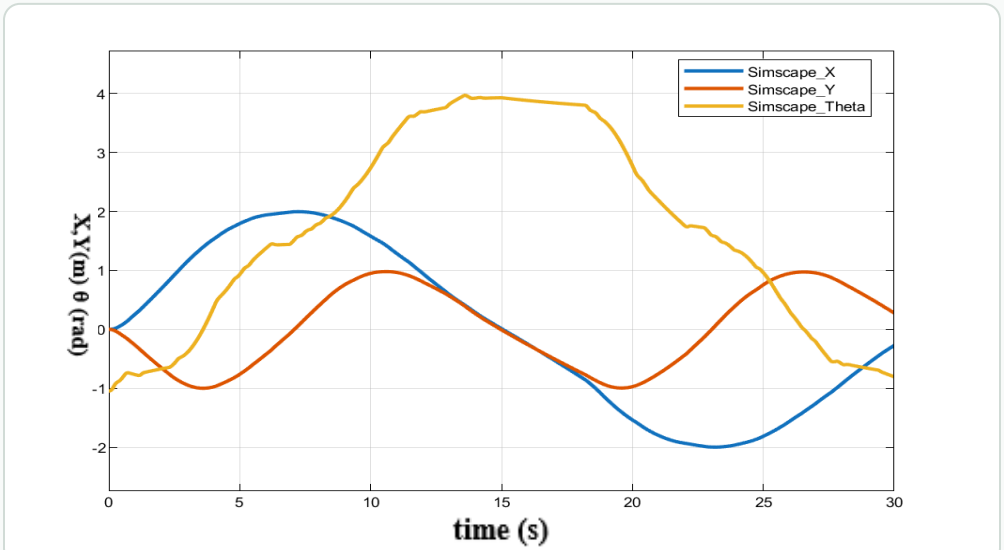
Step 5: H-Bridge & Battery Integration



Hardware Retuning: PID forced down to 0.30 m/s; State-Space kept 0.40 m/s with poles shifted to [-4, -5].

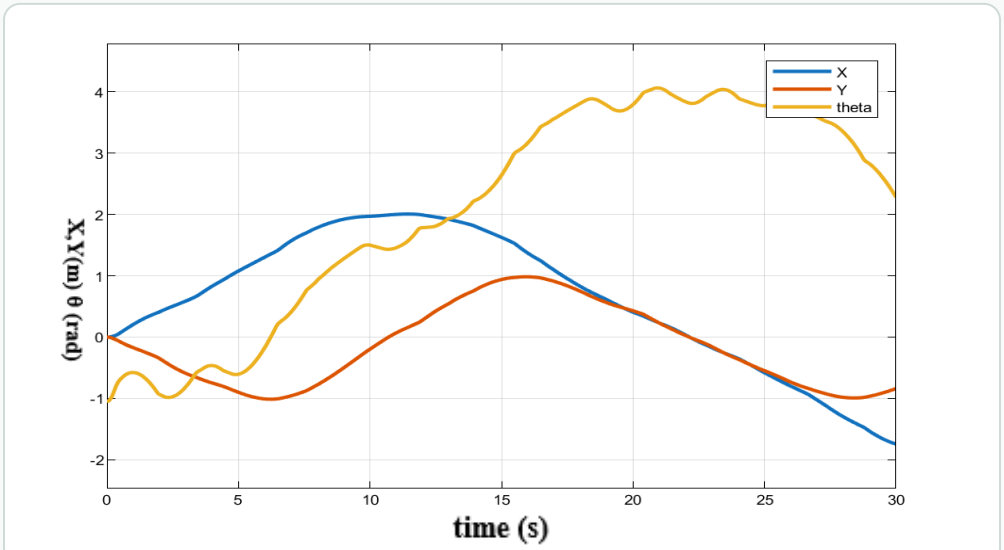
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Final Model Graphs: H-Bridge, Battery & Motor



Final State-Space (0.40 m/s)

Smooth trajectory tracking; travels farther in 30 s with poles [−4, −5].

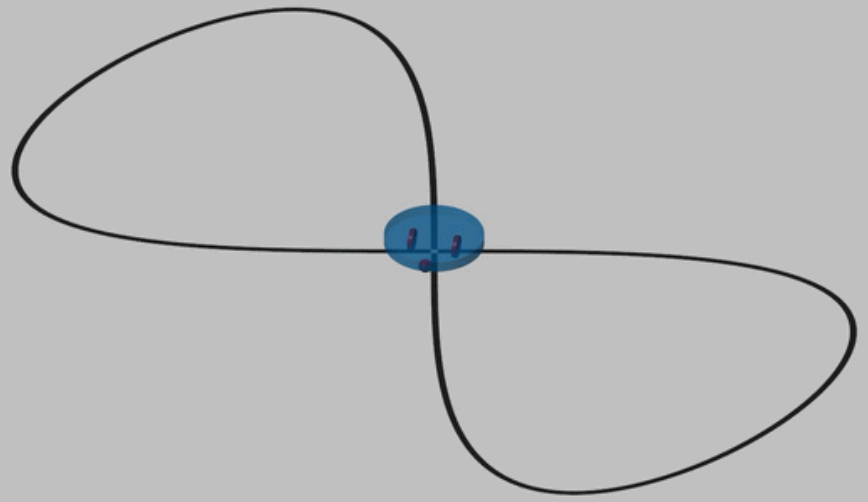


Final PID (0.30 m/s)

Significant wobble and chatter despite reduced 0.30 m/s speed.

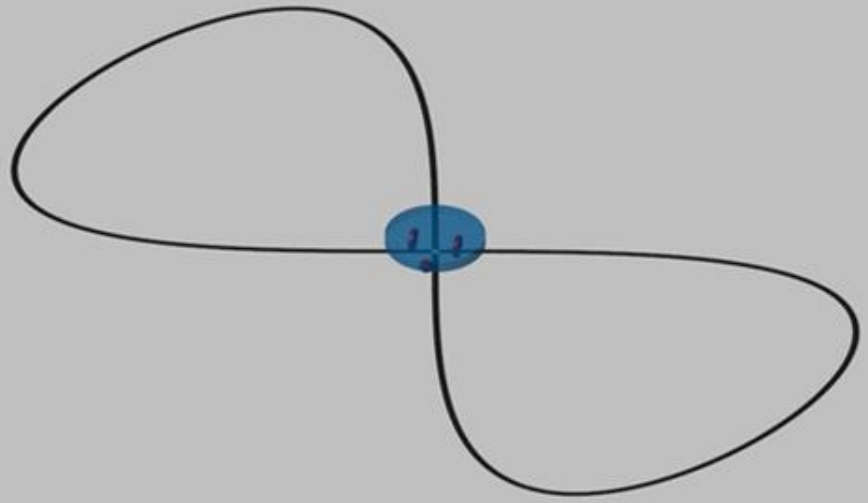
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Video Demonstration: Final Hardware Models



Final PID Hardware Simulation

Reduced to 0.30 m/s (1.08 km/h) with visible trajectory wobble.



Final State-Space Hardware Simulation

Maintains full 0.40 m/s (1.44 km/h) with smooth path tracking.

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Conclusions & Proposed Solutions

Simulation vs. Reality

Ideal models overestimate classical PID stability.

State-Space Robustness

Full-state feedback compensates electromechanical lag.

Ziegler-Nichols & Pole Tuning

Optimize PID and State-Space gains numerically.

Wheel Slip scenarios

Add more realism by including some friction lacking surfaces

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SECTION 7 • APPRECIATION & ACKNOWLEDGMENT

Thank You for Your Attention!

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